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Cryptography and network security

Model Practical

SET-10:-

1) write a c program to Hill cipher succumbs to a known plaintext attack if sufficient plain text - ciphertext pairs are provided. it is even easier to solve the Hill cipher if a choosen pla attack can be mounted.

2) write a c program for subkey generation in c Thefirst subkey is derived from the resulting string by a left shift of one bit and conditio by XORing constant that depend on block size

3) write a c program for perform a letter fre attack on monoalphabetic substitution ci without human intervention.

4) write a c program for Deffie - Hellman

1)

Hill cipher:

AIM:

Analyze and execute the ~~known plain text value~~ Hill cipher with known plaintext value and the cipher text pairs are provided.

Algorithm:-

=> START
=> Read the 2x2 matrix
=> Read the plain text.
=> Multiply the key matrix with plain text
=> Display cipher text
=> STOP.

coding:

```
#include<stdio.h>
int main()
{
int key[2][2], text[2], cipher[2], i, j;
Printf ("Enter 2x2 Matrix:");
for (i=0; i<2; i++)
{
  for (j=0; j<2; j++)
  {
      scanf("%d", & key[i][j]);
  }
}
```

the output was verified.

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```
Printf ("Enter the two plaintext value:");
Scanf ("%d""%d", &text[0], &text[1]);
for (i=0; i<2; i++)
{
  cipher[i] = 0;
}
for (j=0; j<2; j++)
{
  cipher[i] += key[i][j] * text[2];
}
cipher[i] %= 26;
for (i=0; i<2; i++)
{
  Printf ("cipher text", cipher[i]);
}
return 0;
}
```

Input: Enter 2x2 matrix: $\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$

Enter plaintext value: 7, 8

Output:

cipher text : 19 2

result:

Thus the program has been executed successfully and the output was verified.

2) Subkey Generation in CMAC.

AIM:

To generate the subkey in CMAC. first subkey is derived from the resulting string by left shift of one bit.

Algorithm:-

=> START
=> Read L value
=> Left shift L can be key generation of k1.
=> The most significant bit k1
=> Left shift k1 can be using key generation of k2.
=> The most significant is k2.
=> Display k1 and k2.
=> STOP.

Coding:

```
#include <stdio.h>
int main()
{
 signed long long L, k1, k2;
 signed long long Rb: 0x1B;
 Printf ("Enter L value:")
 scanf ("%llx", &L);
```

```
if (L = k1 << 1)
{
   Rb = (L << (1ULL << 63));
}
if (k1 = k2 << 1)
{
   Rb = (k1 << (1ULL << 63));
}
Printf ("%llx", k1);
Printf ("%llx", "The k2 is:", k2);
return 0;
}
```

Input:

Enter L value : 2

output:

The k1 is : 4
The k2 is : 8

result:

Thus, the program has been executed successfully and the output was verified.

3) Letter frequency attack.

AIM:-

To perform a letter frequency attack on monoalphabetic substitution cipher.

Algorithm:-

=> START
=> Read the cipher text.
=> Count the frequency of each alphabet letter.
=> Arrange the letters According to their frequency.
=> Generate possible plaintext
=> Display most likely plaintext.
=> STOP.

Coding:-

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
 char text[500];
int freq[26]={0};
int i,j, temp;
char letter[26];
Printf("enter cipher text:");
```

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```
fgets(text, size of (text));
for (i=0; text[i] != '\0', i++);
{
  if (isalpha(text[i]))
  {
     text[i] = (toupper.text[i]);
  }
}
for (i=0; i<26; i++)
     letters[i] = 'A' + i;
for (i=0; i<26; i++)
{
   for (j=i+1; j<26; j++)
   { if (freq[i] < freq[j])
     { temp = freq[i];
       freq[i]=freq[j];
       freq[j] = temp;
       letters[i] = letter[j];
       letter[j]= temp;
     }
   }
}
Printf("\n letter frequency");
for (i=0 ; i<26; i++)
{ if (freq[i]>0)
   printf ("%c : %d", letter[i], freq[i]);
}return 0;
}
```

Input:

Enter ciphertext: KHOOR ZRUOG

output:

Letter frequency analysis:

O : 3
R : 2
K : 1
H : 1
Z : 1
U : 1
G : 1

Result:-

Thus, the program has been executed successfully and the output was verified.

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4) Diffie- Hellman key Exchange:-

AIM:

To perform a Diffie-Hellman key Exchange.

Algorithm:-

⇒ START
⇒ Read public value of q and a:
⇒ Read Alice private key x
⇒ Read Bob private key y.
⇒ Calculate Alice public key $A = a^x \bmod q$
⇒ calculate Bob public key $B = a^y \bmod q$
⇒ calculate shared secret key.
⇒ Display output
⇒ STOP.

Coding:-

```
#include <stdio.h>
long long Power(long long a, long long b, long long mod)
{
 long long result=1;
 while (b>0)
 {
  result= (result*a);
  b--;
 }return result;
}
int main()
{
```

```
long long q,a,x,y;
long g, long A,B,key1,key2;
Printf("Enter prime number q:")
Scanf("%lld", &q);
Printf("Enter public number a:");
Scanf("%lld",&a);
Printf("Alice secret key x:");
Scanf("%lld",&x);
Printf("Bob secret key y:");
Scanf("%d",&y);
A= Power(a,x,q);
B=Power(B,y,q);
key1=Power(B,x,q);
key2=Power(A,y,q);
Printf("Alice shared key:%d",key1);
Printf("Bob shared key:%d",key2);
return0;
}
```

Input:

Enter prime number: 23
Enter public number: 5
Enter Alice secret key: 6
Enter Bob secret key: 15

output:

Alice send: 8
Bob send: 19
Alice shared key: 2
Bob shared key: 2

Result:

Thus, the program has been executed successfully and the output was verified.